

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

April/May 2012

EE304 Industrial Instrumentation

Date: 01.05.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define the following: [07]
a) Transducers b) Error c) Sensors d) Repeatability e) Dead Zone
f) Calibration g) Backlash
- Q - 2 (a) Explain in detail construction and working of LVDT for pressure measurement. [06]
(b) Explain in detail the construction and working of Bimetallic strips for temperature measurement. [06]
(c) A bimetallic sensor formed of aluminum and copper having thickness of 1mm and length of 20mm each welded at a temperature of 40°C. Find out the amount of bending of the bimetallic strip when it is subjected a temperature of 200°C. (Aluminum $2.5 \times 10^{-5} / ^\circ\text{C}$ and copper $1.7 \times 10^{-5} / ^\circ\text{C}$) [02]

OR

- Q - 2 (a) State various techniques for non-contact type temperature measurement and explain any one in detail. [06]
(b) Explain in detail the pressure measurement sensor which can measure both differential as well as absolute pressure. [06]
(c) Define humidity and state the principle of capacitive hygrometer [02]
- Q - 3 (a) Write a short note on magnetic tape recorder [07]
(b) Classify various level measurement techniques [02]
(c) Explain resistive hygrometers [05]

OR

- Q - 3 (a) Write a short note on X-Y Recorders [07]
(b) Define the following: [07]
a) Absolute pressure b) Gauge pressure c) Absolute humidity d) Relative humidity
e) Gauge factor f) Standards g) Temperature

SECTION - II

- Q - 4 (a) Explain construction and working of RTD and explain various compensation techniques employed for the same. [07]
- Q - 5 (a) Explain the air bubbler method for level measurement [06]
(b) Explain the construction and working of ultrasonic flow meter [08]

OR

OR

- Q - 5 (a) Explain the construction and working of electromagnetic flow meter [08]
(b) Enlist the different types of strain gauges and explain unbounded metal strain gauge with neat diagram. [06]
- Q - 6 (a) A strain gauge having a resistance of 100Ω and gauge factor of 2 is connected in series with a ballast resistance of 100Ω across a 12 volt supply. Calculate difference between output voltage when [1] no stress is applied and [2] stress of 140 MN/m^2 is applied. Take modulus of elasticity is 200 GN/m^2 . [05]
(b) Explain load cells in detail [05]
(c) Explain the method of calibration of strain gauge [04]

OR

- Q - 6 (a) List out various electrical type tachometers used for measuring an angular displacement and explain any one in detail [05]
(b) List out various types of accelerometers and explain any one in detail [05]
(c) Explain in detail radiation type level measurement [04]
